

Who Pays for the KV Cache?

Attributing Shared AI Inference Spend Across Kubernetes and LLM Provider Bills

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github.com/timurista/unalloc · September 2026

Abstract. Organizations now pay for AI through two disconnected ledgers: Kubernetes allocations for self-hosted inference, and per-token bills from API providers, usually through a gateway. We present **unalloc**, an open-source tool that normalizes OpenCost, LiteLLM, OpenAI and Anthropic cost data into one Decimal-exact ledger and reports the share of spend with no owner. We then stress it with five case studies that run inference for real or simulate it faithfully: a discrete-event vLLM-style engine with paged KV memory and prefix caching; a from-scratch PyTorch transformer serving a multi-tenant trace with a real KV cache; tensor- and pipeline-parallel inference on **torch.distributed**; the unmodified CLI against live mock provider APIs; and four downstream use cases. The studies show that the metering rule, not the workload, decides who pays: per-token showback misprices a retrieval-heavy tenant by **33 percentage points** of a pod’s bill relative to measured compute. Distributed serving leaks attribution through pod templates: owner labels set only on leader pods leave **66%** of a LeaderWorkerSet’s GPU bill unowned, and the natural fallback silently assigns **61%** to a Helm chart name. Joining ledgers naively double counts every gateway dollar, and reading one page of a billing API reports a quarter of spend. The studies also surfaced nine defects in the tool itself, all fixed. Code, data, figures and this paper regenerate from one command.

1 Introduction

The cost of an AI feature is split across systems that were never designed to agree. A retrieval-augmented answer might touch a vector database and an embedding job on Kubernetes, a self-hosted open-weights model on a GPU pool, and a frontier model behind an API gateway. OpenCost [11] can price the first two; the gateway and the provider’s billing API can price the last. None of them can say what the feature costs, and none can say what share of the month’s spend belongs to no one — the number FinOps allocation practice is built around [14].

unalloc exists to produce that number. Its design is deliberately small (§2): every source is parsed into one **CostRow** type, labels are canonicalized so **label_costCenter**, **team_id** and a **team:platform** request tag join, and attribution groups the joined ledger on a chosen owner key. The interesting questions are empirical. Shared inference servers batch requests from many tenants [2], share KV memory through paged allocation and prefix caching [1, 4], and scale across devices with tensor and pipeline parallelism [5, 6]. Each of these mechanisms moves cost between tenants, and each leaves a trace — or fails to — in the ledgers a tool like **unalloc** reads.

We ask four questions, each answered by a case study:

1. **Metering.** When one inference pod serves several tenants, how much does the choice of meter (tokens, list-price tokens, compute time, KV memory) change each tenant’s bill? (§4, §5)
2. **Distribution.** What happens to attribution when one model replica spans several pods? (§6)
3. **Joining.** What goes wrong when gateway and provider ledgers are combined? (§7)
4. **Use.** What can an organization do with a joined ledger beyond a percentage? (§8)

Contributions. (i) An open-source ledger-joining tool with exact money handling, deterministic label canonicalization, invoice reconciliation, fallback-aware reporting and a CI budget gate. (ii) Five reproducible case studies — two with real PyTorch inference — whose payloads flow through the tool’s production adapters. (iii) Quantified failure modes of common showback practice. (iv) A catalogue of nine defects the studies exposed in the tool (Table 3), each now covered by a regression test.

2 The joined ledger

2.1 Model

A **CostRow** is one unit of spend: a source, a cost-object name, an amount in USD as a **Decimal**, a time window, normalized labels, and an optional quantity (tokens, GPU-hours). Adapters only parse; they never aggregate or decide attribution. Given rows R , an owner key d and an ordered fallback list $f_1, \dots, f_k$, attribution assigns each row the first non-empty value of $d, f_1, \dots, f_k$, or **<unallocated>**:

$$U(R, d) = \frac{\sum_{r \in R, \text{owner}(r)=\perp} \text{amount}(r)}{\sum_{r \in R} \text{amount}(r)}$$

Because fallbacks make the headline smaller without making it more correct, **unalloc** also reports F , the spend attributed **only** through a fallback key (§6 shows why this matters).

2.2 Canonicalization

Label keys are lower-cased, camelCase is split, path-style keys keep their last segment (**app.kubernetes.io/name** $\rightarrow$ **name**), known provider prefixes are stripped until none remain, and an alias table maps spellings (**team_id**, **owner**, **squad**) onto one dimension. When distinct raw keys collapse onto one canonical key, a fixed precedence decides: a key already written canonically, then an alias, then a rewritten key, then a path-derived key, with ties broken on the raw key. Before the case studies the tool kept whichever key arrived first, which made results depend on provider serialization order.

2.3 Reconciliation and gates

reconcile compares each source’s ledger total with a billed amount and flags deltas above a tolerance; **report --budget P** exits non-zero when $U > P$, so a deployment pipeline can refuse to ship unowned spend.

3 Methodology

Table 1 summarizes the studies. Every study emits payloads in the providers’ own response shapes and parses them with **unalloc**’s production adapters; the end-to-end study goes further and runs the CLI binary against HTTP servers. Prices are deliberately round: one H100-class GPU at \$3.00/hour, OpenCost’s default CPU and RAM rates, and illustrative per-million-token API prices. The questions are about **shares**, which do not depend on the absolute rates. Unless stated, a monthly pool is $8 \text{ GPUs} \times 720 \text{ h} = \$17,280$.

Study	What runs	Real vs. modelled	Runtime
kv_cache	Discrete-event vLLM-style engine: paged KV blocks, prefix caching, chunked prefill, continuous batching, recompute preemption; 4 tenants, Poisson + multi-turn sessions	modelled latency	102 s
torch_kv	3.3M-parameter decoder (4 layers, $d = 256$, RoPE [9]) with explicit KV cache; 96-request 4-tenant trace on CPU	real inference	29 s
distributed	Megatron-style TP (2, 4 ranks) and 2-stage PP on torch.distributed/gloo ; LeaderWorkerSet cost scenarios	real inference	24 s
hybrid_e2e	CLI subprocess vs. mock OpenCost, LiteLLM, OpenAI, Anthropic APIs with real auth schemes and cursor pagination	real CLI, synthetic spend	2 s
use_cases	Labeling Pareto, feature unit economics, self-host break-even, CI gate	mixed	25 s

Table 1: Case studies. Runtimes on an Apple-silicon laptop (10 cores), Python 3.11, PyTorch 2.14 CPU.

4 A shared vLLM pod: the meter decides who pays

4.1 Setup

The simulator models one replica of an 8-replica fleet serving an 8B-parameter model (131 KB of KV per token) from a 12,000-block pool (16 tokens per block). Engine steps share a 2,048-token budget between decodes and chunked prefills [3], and step latency is an analytic function of prefill tokens, decode sequences

and attended context. Four tenants arrive by Poisson process: **search** (RAG: 600–1,400-token prompts, 16–64-token answers, three shared system prompts), **agents** (multi-turn sessions that re-send the whole conversation each turn, 150–400-token outputs), **platform** (medium/medium) and an unlabeled shared **sandbox** key. Each step’s duration is split across the sequences in it by their marginal work; each KV block-second is split across the requests holding the block. Idle time and unheld blocks are recorded separately as overhead.

At 3 requests/s for 30 minutes the engine served 10,020 requests with a 75% prefix-cache hit rate, no preemptions, and a median time-to-first-token of 38–96 ms per tenant. (At 4 requests/s a five-minute run looks healthy but agent follow-up turns accumulate over half an hour and median TTFT exceeds 40 s; we report the sustainable operating point.)

4.2 Results

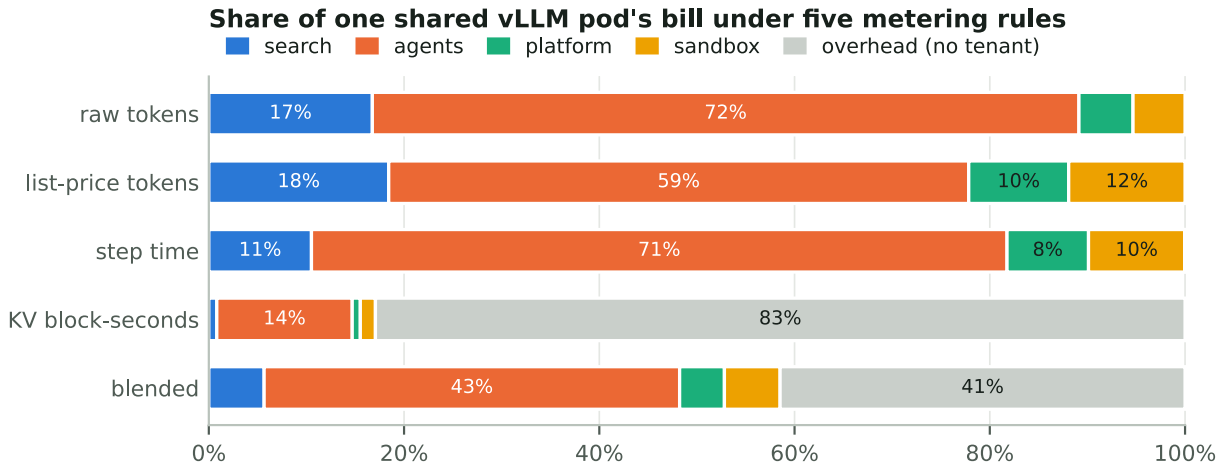


Figure 1: One pod bill, five meters. Token meters have no notion of overhead. Step-time metering sees none either once continuous batching keeps any request in flight; KV-memory metering leaves 83% of the bill with no request.

Figure 1 shows the split. Agents dominate every compute-like meter because each turn re-sends a growing conversation, but the meters still disagree: raw tokens charge **search** 16.7% of the pod, step time 10.5% — a 6.2-point gap that grows to 12.0 points against KV memory once overhead is redistributed. List-price weighting (cached input at 0.1 $\times$, output at 4 $\times$) is no closer: it moves agents from 72.4% to 59.4%, twelve points away from measured step time (71.2%).

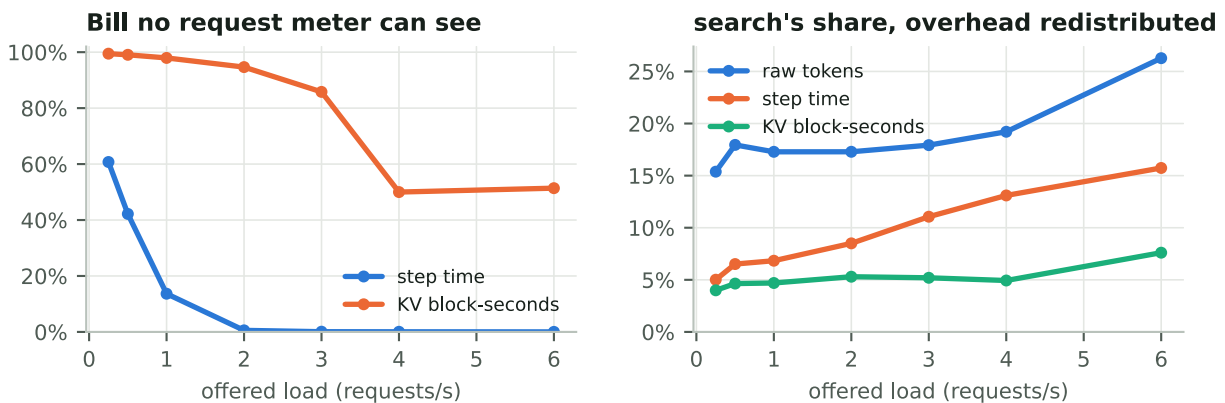


Figure 2: Left: overhead visible to each meter as load rises. Right: search’s share under three meters, overhead redistributed.

Figure 2 explains why. Step-time overhead collapses to 0 by 2 requests/s: a pod that is “busy” in wall-clock terms is nowhere near its token capacity (754 vs. 1,340 output tokens/s). KV memory tells the opposite story — 95% of block-seconds are unheld at the same load, because pools are provisioned for peaks. Neither is wrong; they measure different scarce resources, and the choice moves several points of the bill between tenants.

For unalloc. A per-tenant split only closes the gap if overhead is labeled. Emitting the memory split with an unlabeled overhead row reports 84.5% unallocated; redistributing the same overhead as OpenCost **sharedCost** reports 9.1% — only the sandbox key. The unallocated percentage is as much a statement about the overhead policy as about labels.

5 Real inference with a KV cache

5.1 Setup

A from-scratch decoder-only transformer (3.28M parameters, random weights with a fixed seed — serving cost does not depend on weight values) implements a preallocated per-layer KV cache and shared-prefix reuse. Before any timing, cached decoding is checked against full recomputation: generated tokens are identical and the maximum logit difference is 1.07×10^{-6} . The trace has 96 requests from the same four tenant shapes; it is served three times and per-request medians are used. Measured shares moved by at most 4.3 points across repeats.

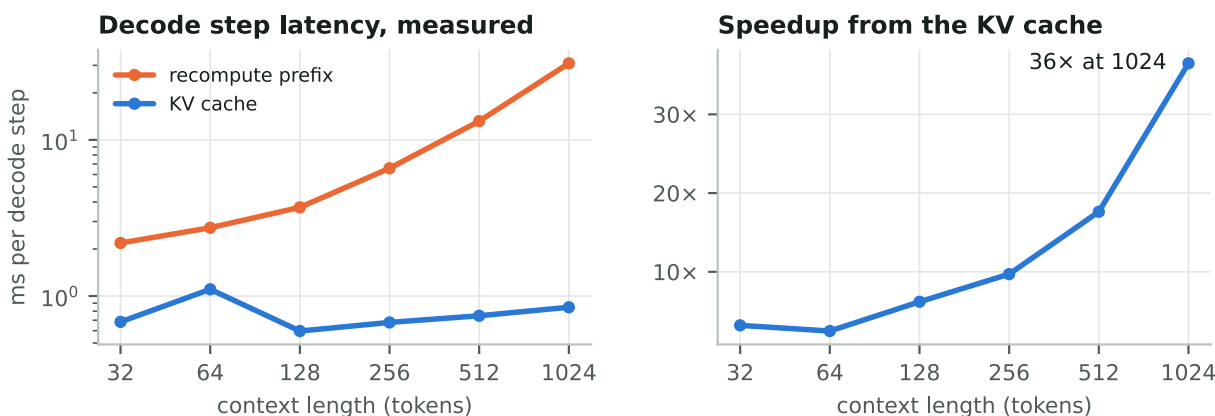


Figure 3: Measured decode-step latency on CPU. At 1,024 tokens of context the cache is 8.0 MiB and a step takes 0.85 ms instead of 30.9 ms.

5.2 Results

Figure 3 reproduces the expected scaling [7]: recomputation grows with context, cached steps stay flat, 36.5× faster at 1,024 tokens. The attribution consequences are in Figure 4. Search sent 18,024 prompt tokens and received 701; agents sent 2,345 and received 4,017. Token counting (and analytic FLOPs [8], which track tokens) charge search 45% of the pool. Measured compute charges it 12.4%: prefill is batched and cheap, decode is sequential and expensive.

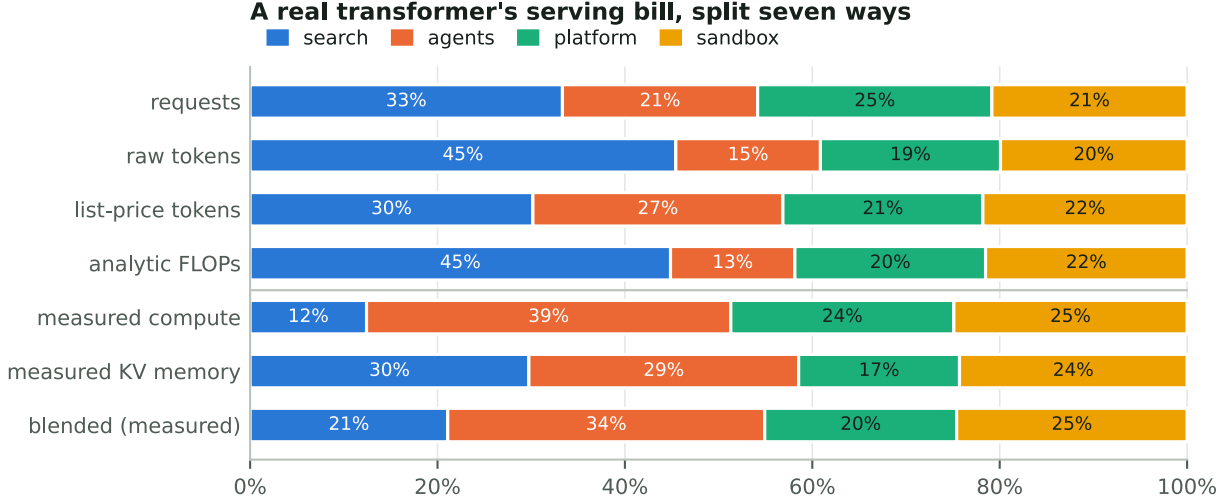


Figure 4: The same measured trace split seven ways. Above the rule: meters derivable from request logs. Below: meters that require serving telemetry.

Headline. Per-token showback overcharges the RAG tenant by **33.0 points** of the pool — \$5,703/month on \$17,280 — relative to measured compute. Measured cost per token differs $9.2\times$ between search and agents; a flat token price charges them identically. Pricing output at $4\times$ halves the gap (17.8 points) without closing it.

On a GPU the per-step fixed overhead is amortized across a batch, so the magnitude is hardware-specific; the direction — prompt-heavy tenants subsidize decode-heavy ones under token pricing — is structural, and matches the prefill/decode asymmetry that motivates disaggregated serving [10].

6 Distributed inference: attribution leaks through pod templates

6.1 Parallel inference, verified

We shard the reference model Megatron-style [5] — column-parallel QKV and MLP up-projection, row-parallel output and down-projection with an all-reduce after each — and separately split it into two pipeline stages that pass activations with send/recv [6]. All configurations generate the same tokens as the single-process model (maximum logit difference 3.3×10^{-6} for TP, exactly 0 for PP). Per-rank parameters drop from 13.65 MB to 7.36 MB (TP2) and 4.21 MB (TP4); embeddings, norms and the LM head stay replicated. KV cache per rank halves with each doubling.

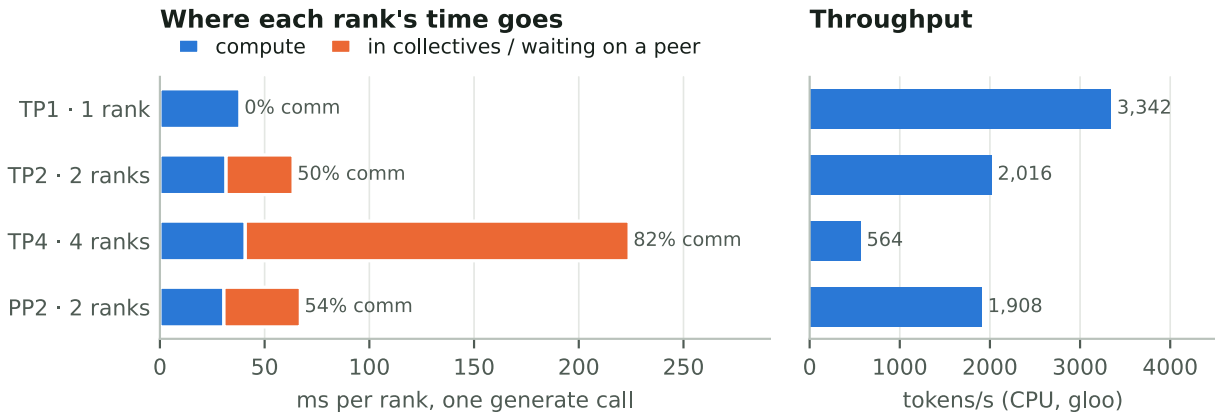


Figure 5: Per-rank time split for batch 4, 64-token prompt, 32 new tokens. Communication is time blocked in collectives or waiting for a peer.

On CPU over loopback, distribution is slower, as expected (Figure 5): TP4 spends 82% of each rank’s time in collectives. These fractions are an upper bound for NVLink-connected GPUs; we use them only to size the scenario below and report a sensitivity series at 5–20%.

6.2 The LeaderWorkerSet month

Multi-host serving on Kubernetes commonly runs one replica as a group of pods, e.g. with LeaderWorkerSet [13]: a leader template and a worker template. We emit one month of OpenCost allocations for two tenants — **search** (2 replicas $\times$ 4 pods, tensor parallel) and **agents** (3 replicas $\times$ 2 pods, pipeline parallel) — plus cluster idle, \$38,400 in total, with each pod’s GPU time split into compute, communication and in-pod idle using the measured fractions.

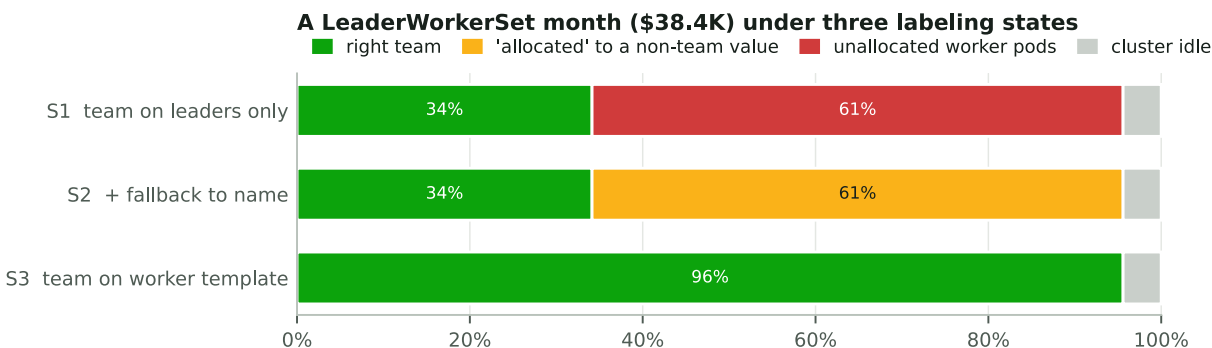


Figure 6: Correctness of attribution, not just coverage, under three labeling states.

In S1 the **team** label is on the leader template only: 65.9% of the month is unallocated. The obvious rescue is a fallback key every pod carries. In S2, falling back to **name** drops the unallocated share to 4.4% — and routes \$23,597 (61%) into a bucket called `vllm`, the Helm chart’s `app.kubernetes.io/name`, which collides with `leaderworkerset.sigs.k8s.io/name` on the canonical key `name`. S3 fixes the worker template: 95.6% correct, only idle remains.

For unalloc. The headline percentage cannot distinguish S2 from S3. We added the fallback-attributed amount F to every report (\$23,597 in S2, \$0 in S3) and made collision resolution order-independent; before the fix the S2 bucket was `vllm` or `search-llama-70b` depending on JSON key order.

Communication is itself a cost with an owner. GPU time spent in collectives totals \$12,059, \$8,201 of it on worker pods that S1 leaves unowned. A per-token showback, which spreads it by token share, moves \$3,737 from the tensor-parallel tenant onto the pipeline-parallel one; at a more GPU-like 10% communication fraction the transfer is \$166.

7 Joining gateway and provider ledgers

The organization in this study runs a LiteLLM gateway [12] in front of OpenAI and Anthropic, the self-hosted fleet above, and — like most — a research team calling providers directly. Mock servers enforce each provider’s authentication (Bearer for OpenAI and LiteLLM, `x-api-key` for Anthropic’s Admin API) and page responses with cursors; in the dev container the LiteLLM spend log lives in Postgres, as it does in production. The CLI runs as a subprocess with only environment variables pointing it at the servers.

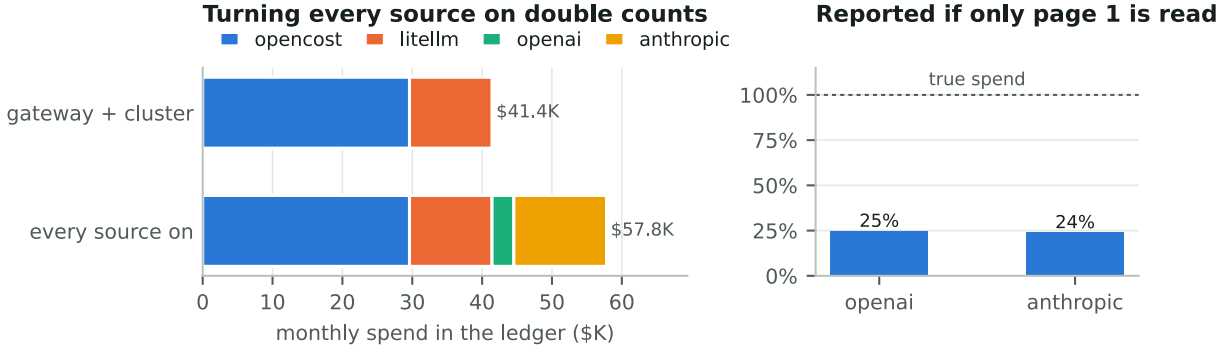


Figure 7: Left: ledger totals for the recommended sources versus every source. Right: share of true provider spend reported by an adapter that reads only the first page.

Scenario	Ledger total	Unallocated
A. Cluster + gateway (<code>--source opencost --source litellm</code>)	\$41,420	66.9%
B. Every source enabled	\$57,813	76.3%
C. Provider billing only	\$16,393	100.0%
C'. Provider billing, fallback to project/workspace	\$16,393	0.0%
Provider ledgers vs. provider invoices	matched	0.0% delta
Gateway ledger vs. provider invoices	\$11,815 of \$16,393	-27.9%

Table 2: End-to-end scenarios over one month.

Table 2 and Figure 7 give three results. **Double counting**: enabling every source adds \$16,393, of which \$11,815 — exactly the gateway’s spend — is counted twice; the remainder is real bypass traffic the gateway never saw. **Coverage**: the gateway ledger reconciles to only 72.1% of the provider invoices, so a gateway-only view under-reports by \$4,578 and cannot know it without reconciliation. **Fallback illusion**: provider billing alone has no team dimension; falling back to project and workspace reports 0% unallocated while attributing nothing to a team.

Before the fixes in Table 3, this study could not run at all: the OpenAI and Anthropic adapters ignored their default URLs whenever the CLI passed an unset environment variable, Anthropic rejected the Bearer header, and neither adapter followed pagination — reading 25% of OpenAI spend and 24% of Anthropic spend.

8 Use cases for a joined ledger

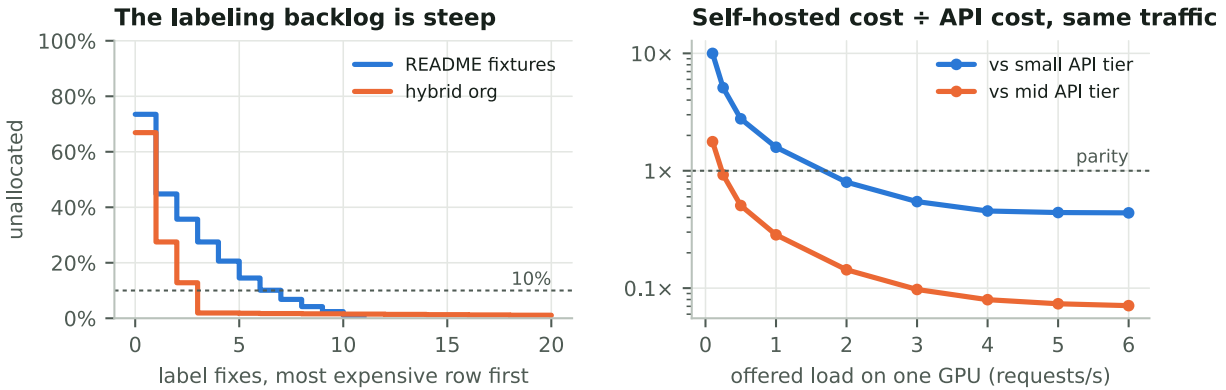


Figure 8: Left: unallocated share after fixing the most expensive unlabeled rows first. Right: cost of serving the simulated traffic on one self-hosted GPU divided by the same tokens at API list prices, for two illustrative price tiers.

Labeling is a Pareto problem. Ordering the backlog by dollars (Figure 8, left), three label changes take the hybrid organization from 66.9% to 1.9% unallocated; the README fixtures need seven changes to cross 10%. A dollar-ordered backlog turns a governance program into a short ticket list.

Feature unit economics need both ledgers. Joining on a **feature** label, the RAG **answer** feature costs \$22.32 per thousand requests including its vector database, against \$13.38 if only the LLM bill is counted: 40% of the feature’s cost is invisible to a gateway-only view.

Build-versus-buy depends on utilization, not list price. One self-hosted GPU serving the simulated traffic is cheaper than a mid-tier API above 0.23 requests/s and cheaper than a small-tier API above 1.7 requests/s (Figure 8, right), flattening at 0.08× and 0.45× respectively once saturated. The comparison bounds cost, not quality.

Budgets belong in CI. `unalloc report --budget 50` exits 2 on the fixtures (73.5% unallocated); `--budget 80` exits 0, so a deployment that ships unlabeled workloads can fail review.

Further use cases. The same ledger supports multi-LoRA serving (attribution by adapter label on a shared base model), fine-tuning and batch embedding jobs whose GPU hours share clusters with online inference, agent systems whose tool calls fan out across providers, speculative decoding where a draft model’s cost belongs to the target model’s callers, and multi-cloud estates where Bedrock and Azure OpenAI bills join the same dimension.

9 Defects found by the case studies

Area	Defect	Found by
Fetch	OpenAI/Anthropic default base URL overridden by an unset env var; every live fetch failed	review; e2e
Fetch	Anthropic Admin API called with Bearer auth instead of <code>x-api-key + anthropic-version</code>	review; e2e
Fetch	No cursor pagination on provider cost APIs; first page only (25% of spend)	review; e2e
Money	OpenCost component costs summed through <code>float</code> when <code>totalCost</code> was absent	review
Labels	Only one provider prefix stripped; <code>label_app_kubernetes_io_name</code> $\neq$ <code>app.kubernetes.io/name</code>	distributed
Labels	Canonical-key collisions resolved by arrival order, incl. explicit key vs. alias	distributed, e2e
Labels	OpenCost annotations silently overrode same-key labels	distributed
Report	<code>labels</code> backlog ignored <code>--fallback</code> , disagreeing with <code>report</code>	review
Report	<code>labels</code> printed “Nothing unallocated” when no source loaded	review

Table 3: Defects fixed in the course of the studies. Each has a regression test.

10 Discussion and limitations

Recommendations. (1) Decide the meter explicitly and publish it; token counts are a pricing choice, not a measurement. (2) Keep overhead — idle GPUs, empty KV pools, communication — as its own labeled line and redistribute it by policy, rather than letting any meter absorb it silently. (3) Put owner labels on every pod template of a multi-pod workload, and treat fallback-attributed spend as unverified. (4) Choose sources so each dollar has one path into the ledger, and reconcile against invoices every month.

Limitations. PyTorch timings are from CPU, where fixed per-step overhead and loopback collectives inflate decode and communication costs; the directions we report are structural but the magnitudes are not GPU numbers. The serving simulator uses analytic step latency and synthetic workloads. Dollar figures use round, illustrative prices. The mock provider APIs reproduce authentication and pagination semantics as documented, not every field of the live responses.

11 Reproducibility

Everything in this paper regenerates from the repository:

```
make research          # CPU PyTorch, notebook tooling, typst
make case-studies      # writes case_studies/results/*/metrics.json
make paper             # figures + this PDF
```

```
make notebook          # executes the companion notebook
make devcontainer-check # the same, inside the isolated dev container
```

A browser-based ledger explorer (`python -m case_studies.ui`) shows every dataset row by row with the owner each row resolves to, and a companion notebook walks through each study.

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